Breaking SAM 2: An Evaluative Benchmark on Extreme Object Evolution

1. Goal and Research Questions

This project will systematically identify, quantify, and explain the failure modes of zero-shot video object segmentation (VOS), focusing on SAM 2 and the Grounded-SAM-2 pipeline. The central hypothesis is that temporal propagation becomes unreliable when object identity is not represented by stable appearance and shape cues. Rather than asking only whether tracking succeeds, the benchmark will determine when, how, and under which transformation it fails.

- **Topology and state change:** highly deformable or non-solid targets such as pouring liquids, flames, smoke, melting material, or kneaded dough, where splitting, merging, and changing boundaries violate conventional object permanence assumptions.
- **Extreme scale and viewpoint:** objects rapidly approaching or receding from the camera, producing large changes in visible area, texture resolution, and partial visibility.
- **Abrupt illumination:** sudden shadows, lights switching off, tunnel entry, glare, or exposure changes that alter appearance without changing semantic identity.

Key questions are: (i) which disruption produces the earliest and most severe degradation; (ii) whether errors are abrupt, gradual, or recoverable; (iii) how sensitive results are to the initial prompt; and (iv) whether text-guided initialization improves localization or merely introduces an additional first-frame detection failure.

2. Prior Work and Motivation

Grounding DINO is an open-set detector that tightly fuses language and image features to localize objects described by category names or referring expressions. SAM 2 extends promptable segmentation to video through a transformer with streaming memory, conditioning each frame on prompts and stored information from earlier frames. Grounded-SAM-2 combines these components in a modular chain: text prompt → Grounding DINO box → SAM 2 mask → temporal mask propagation.

This composition is powerful but creates two distinct error sources. First, the detector may select the wrong instance or an incomplete box on the initial frame. Second, even with a correct initialization, SAM 2 may accumulate temporal drift when appearance, topology, occlusion, or scale changes invalidate previously stored cues. The project will therefore separate initialization quality from propagation quality instead of treating the pipeline as a single black box.

DAVIS and YouTube-VOS established region similarity (Jaccard/IoU, J) and contour accuracy (boundary F-measure, F) as standard VOS metrics. However, broad benchmarks emphasize overall averages across common object categories. A compact stress-test benchmark can complement them by concentrating difficult transformations and reporting temporal failure behavior, not only mean accuracy.

3. Methods

- **Dataset curation:** Compile 15–30 short clips (approximately 5–10 seconds) from reusable/open sources such as Pexels and appropriately licensed videos. Each clip will receive a primary disruption label, target description, frame rate, resolution, and event timestamp. Clips with ambiguous ownership, severe compression, or unclear target identity will be excluded.
- **Ground truth:** Annotate the target mask on every 5th–10th frame using CVAT, with denser annotation around the transformation event. Keyframe-only evaluation avoids an excessive labeling burden while preserving the temporal pattern of failure. A small subset may be densely annotated to confirm that sparse sampling does not hide short failures.
- **Pipelines:** A: SAM 2 initialized with a manually verified box or click on frame 1. B: Grounded-SAM-2 initialized from a fixed text prompt. An optional oracle-box variant will feed Grounding DINO's selected object to SAM 2 only after manual box verification, isolating detection from propagation errors.
- **Metrics:** Report mean J, F, and J&F; per-category performance; worst-frame accuracy; temporal stability; and failure-frame statistics. Define the first failure as the first annotated frame below a predeclared J threshold (e.g., 0.5) for two consecutive observations. Also record recovery rate, mask-area drift, fragmentation, and false-positive persistence after the target disappears.
- **Analysis:** Plot accuracy against normalized video time and align clips by the disruption event. Qualitative labels will distinguish target loss, background takeover, identity switch, mask fragmentation, boundary lag, and over-segmentation. Results will be stratified by disruption type and initialization method.

4. Resources and Computing Requirements

The project is inference-only; no fine-tuning is required. Development can be performed locally for pipeline integration and small tests, while final evaluation can use a single consumer or cloud GPU (for example, an RTX-class GPU or Colab T4/L4). The software stack will use PyTorch, the official SAM 2 and Grounding DINO implementations, OpenCV/FFmpeg for video processing, CVAT for annotation, and Python tools for metric computation. Expected storage, including clips, masks, predictions, and model weights, is below approximately 15 GB.

5. Challenges, Mitigation, and Deliverables

- **Annotation bottleneck:** Limit the benchmark size, annotate sparsely, densify only near critical events, and use model-assisted masks followed by manual correction rather than drawing every mask from scratch.
- **Subjective failure definitions:** Pre-register the IoU threshold and consecutive-frame rule; report continuous J/F curves alongside the binary failure frame; and have a second reviewer inspect borderline cases.
- **Unexpected robustness or dataset bias:** Pilot a larger candidate pool, retain clips according to explicit inclusion criteria rather than cherry-picking only failures, and include several control clips with ordinary motion to verify that the pipeline is configured correctly.

The final deliverables will be: (1) a documented Evolution Benchmark with annotations and metadata; (2) reproducible inference and evaluation code; (3) quantitative tables and temporal failure plots; and (4) a concise taxonomy of failure modes with representative visual examples. Because the work avoids model training, the primary risks are controllable data and evaluation tasks, making the project feasible within a semester while still producing a rigorous and potentially reusable benchmark.

Selected references: [1] Ravi et al., "SAM 2: Segment Anything in Images and Videos," 2024. [2] Liu et al., "Grounding DINO," 2023. [3] Perazzi et al., DAVIS benchmark, 2016. [4] Xu et al., YouTube-VOS, 2018.